FINAL REPORT

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INTELLIGENT CONSTRUCTION SITE SAFETY MONITORING SYSTEM

1.Introduction

Construction sites are widely recognized as one of the most hazardous working environments, where failure to comply with safety protocols—particularly the use of Personal Protective Equipment (PPE)—can result in severe injuries or fatalities. Ensuring worker safety is therefore a critical responsibility; however, traditional monitoring methods rely heavily on manual supervision, which is often inefficient, inconsistent, and difficult to scale.

Key limitations of traditional monitoring include:

- Dependence on human observation
- Inconsistent supervision due to fatigue
- Limited scalability across large sites
- Delayed response to unsafe behaviors

With the advancement of artificial intelligence and computer vision, automated monitoring systems have emerged as a reliable solution. This project introduces an intelligent system that integrates deep learning, tracking, and rule-based reasoning to detect PPE compliance in real time.

Unlike conventional approaches, the proposed system incorporates:

- Temporal reasoning across frames
- Violation persistence logic
- Multi-stage validation mechanisms

These features enable reliable operation under real-world conditions such as occlusion and motion blur.

2.Problem Statement

Manual PPE monitoring presents several operational and technical challenges.

Operational Challenges

- Human fatigue leading to missed violations
- Difficulty monitoring multiple workers simultaneously
- Lack of continuous tracking of worker behavior
- No structured historical safety records

Technical Challenges in Existing AI Systems

- Frame-level detection inconsistency
- Difficulty detecting absence of PPE (e.g., no helmet)
- Sensitivity to lighting and occlusion
- Dataset bias affecting generalization

To address these issues, this project proposes a system that combines:

- Object detection
- Multi-object tracking
- Temporal analysis
- Rule-based decision-making

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3. System Overview

The system follows a modular pipeline architecture designed for robustness and scalability.

YOLOv8 Detection → ByteTrack Tracking → PPE Matching → Safety Rules Engine → Alert Generation → Visualization → Analytics → Incident Review

Core Components

The system consists of the following major modules:

- **YOLOv8 Detection Model** – identifies workers and PPE
- **ByteTrack Tracking Module** – maintains worker identity across frames
- **PPE Matching System** – associates PPE with workers
- **Safety Rules Engine** – evaluates compliance over time
- **Alert System** – generates safety notifications
- **Visualization Engine** – displays real-time outputs
- **Analytics Engine** – computes safety metrics
- **Incident Review System** – validates detected violations

This layered design ensures that decisions are based on **consistent temporal observations rather than single-frame predictions**.

4. Dataset Engineering

The performance of the system depends heavily on dataset quality and diversity.

4.1 Primary Dataset

The main dataset consists of:

- Total images: 43,999
- Training images: 30,765
- Validation images: 8,814
- Test images: 4,423

4.2 Dataset Characteristics

The dataset includes:

- ~76,398 bounding box annotations
- Multiple PPE-related classes
- Strong representation of common PPE (helmets, vests)

However, the following issues were identified:

- Class imbalance in certain categories
- Limited representation of non-standard PPE

4.3 Custom Dataset Development

To improve model performance, a custom dataset of 125 images was created.

Motivation

- Improve detection of blue safety vests
- Handle non-standard PPE variations

Data Sources

- Pexels
- Unsplash

Integration Approach

- Class labels mapped consistently
- Dataset merged without disrupting structure

5. Model Development and Training

5.1 Model Configuration

The system uses the YOLOv8m architecture with the following settings:

- Image size: 640
- Confidence threshold: 0.25
- Model: best_merged_ppe_yolov8m.pt

5.2 Training Setup

The model was trained under the following conditions:

- GPU: NVIDIA A100
- Batch size: 16
- Epochs: ~100
- Framework: Ultralytics YOLOv8

5.3 Performance Metrics

Metric	Value
Precision	0.713
Recall	0.852
mAP@50	0.792
mAP@50-95	0.538

5.4 Performance Interpretation

The results indicate:

- High recall → fewer missed safety violations
- Moderate precision → acceptable false positives
- Lower mAP@50-95 → bounding box precision challenges

6.Multi-Object Tracking (ByteTrack)

To enable temporal analysis, the system integrates ByteTrack.

Key Configuration Parameters

- track_high_thresh: 0.5
- track_low_thresh: 0.1
- track_buffer: 60
- match_thresh: 0.8

Advantages of Tracking

- Maintains consistent worker identities
- Enables behavior analysis over time
- Supports violation persistence logic

7. PPE Matching Mechanism

After detection and tracking, PPE items are assigned to workers.

Matching Techniques

- Bounding box overlap
- Spatial proximity

Thresholds Used

- Helmet: 0.05
- Vest: 0.10

This ensures accurate association even in crowded scenes.

8. Safety Rules Engine

The safety rules engine is the core decision-making component.

8.1 Worker Classification

Each worker is classified into:

- Safe → all PPE present
- Unsafe → PPE missing consistently
- Uncertain → insufficient evidence

8.2 Temporal Memory

The system tracks:

- PPE presence history
- Missing durations
- Detection consistency

8.3 Violation Timer

A worker is classified as unsafe only if:

- Violation persists for more than 6 seconds

8.4 Negative Evidence

The system leverages:

- NO-Hardhat detections
- NO-Safety Vest detections

8.5 Risk Scoring

Risk scores are computed based on:

- Duration of violation
- Detection consistency
- PPE absence

9. Alert Generation System

Alerts are generated based on worker safety state.

Alert Levels

- Info → Safe
- Warning → Uncertain
- Critical → Unsafe

Alert Details Include

- Worker ID
- Violation reason
- Supporting notes

10. Visualization and Analytics

10.1 Visualization

The system overlays:

- Bounding boxes
- Color-coded safety states
- Risk scores
- Alerts

10.2 Analytics Metrics

The analytics engine computes:

- Average workers per frame
- Helmet compliance rate
- Vest compliance rate
- Full PPE compliance rate

These metrics assist in decision-making and reporting.

11. Incident Review Mechanism

A secondary validation layer is implemented to:

- Re-evaluate detected violations
- Reduce false positives
- Improve system reliability

12. Limitations

Despite strong performance, several limitations exist:

- False detection of orange/yellow T-shirts as safety vests
- Class imbalance affecting accuracy
- Difficulty detecting absence of PPE
- Reduced performance under occlusion

13. Future Improvements

Future enhancements may include:

- Expanding dataset diversity
- Improving PPE detection using reflective features
- Integrating multi-camera systems
- Deploying on edge devices

14. Conclusion

This project successfully demonstrates the development of an intelligent safety monitoring system for construction environments. By integrating detection, tracking, and rule-based reasoning, the system achieves reliable PPE compliance monitoring in real time.

Key contributions include:

- Integration of temporal reasoning
- Reduction of false positives using violation timers
- Improved robustness under real-world conditions

The system represents a significant improvement over traditional monitoring approaches and provides a scalable solution for enhancing workplace safety.

15.Appendix

Image1:Detection and analysis

```
--------------------------------------------------------------------------------
PROCESSING COMPLETED
--------------------------------------------------------------------------------
Frames processed      : 1111
Unique worker IDs     : 61
Total alerts generated: 9803
Elapsed time          : 0:12:02.802708
--------------------------------------------------------------------------------
LATEST FRAME STATE SUMMARY
Safe: 8 | Unsafe: 0 | Uncertain: 0
Info alerts: 8 | Warning alerts: 0 | Critical alerts: 0
--------------------------------------------------------------------------------
SECOND-LEVEL REVIEW SUMMARY
Likely compliant: 44 | Manual review required: 0 | Sustained likely violation: 0 | Insufficient observation: 1
--------------------------------------------------------------------------------
ANALYTICS SUMMARY
Average persons detected       : 9.69
Average helmet matched         : 8.91
Average vest matched           : 7.91
Average full PPE               : 7.33
Helmet compliance rate         : 91.95%
Vest compliance rate           : 81.67%
Full PPE compliance rate       : 75.65%
--------------------------------------------------------------------------------
OUTPUT FILES
Annotated video       : G:\site-safety-vision\outputs\predictions\test1_annotated.mp4
Inference JSON        : G:\site-safety-vision\outputs\predictions\test1_results.json
Review JSON           : G:\site-safety-vision\outputs\predictions\test1_review.json
Analytics JSON        : G:\site-safety-vision\outputs\predictions\test1_analytics.json
Incident snapshots    : G:\site-safety-vision\outputs\predictions\incidents
--------------------------------------------------------------------------------
(.venv) PS G:\site-safety-vision>
```

Image2:Safety analysis report

```
--------------------------------------------------------------------------------
Processing started...
[Frame 0] workers=8 | safe=7 | unsafe=0 | uncertain=1 | alerts=8
[Frame 30] workers=8 | safe=7 | unsafe=0 | uncertain=1 | alerts=8
[Frame 60] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 90] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 120] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 150] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 180] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 210] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 240] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 270] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 300] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 330] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 360] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 390] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 420] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 450] workers=9 | safe=8 | unsafe=1 | uncertain=0 | alerts=9
[Frame 480] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 510] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 540] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 570] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 600] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 630] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 660] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 690] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 720] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 750] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 780] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 810] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 840] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 870] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
[Frame 900] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 930] workers=6 | safe=6 | unsafe=0 | uncertain=0 | alerts=6
[Frame 960] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 990] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 1020] workers=6 | safe=6 | unsafe=0 | uncertain=0 | alerts=6
[Frame 1050] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
[Frame 1080] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
[Frame 1110] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
--------------------------------------------------------------------------------
```

Image3:Frame by frame analysis

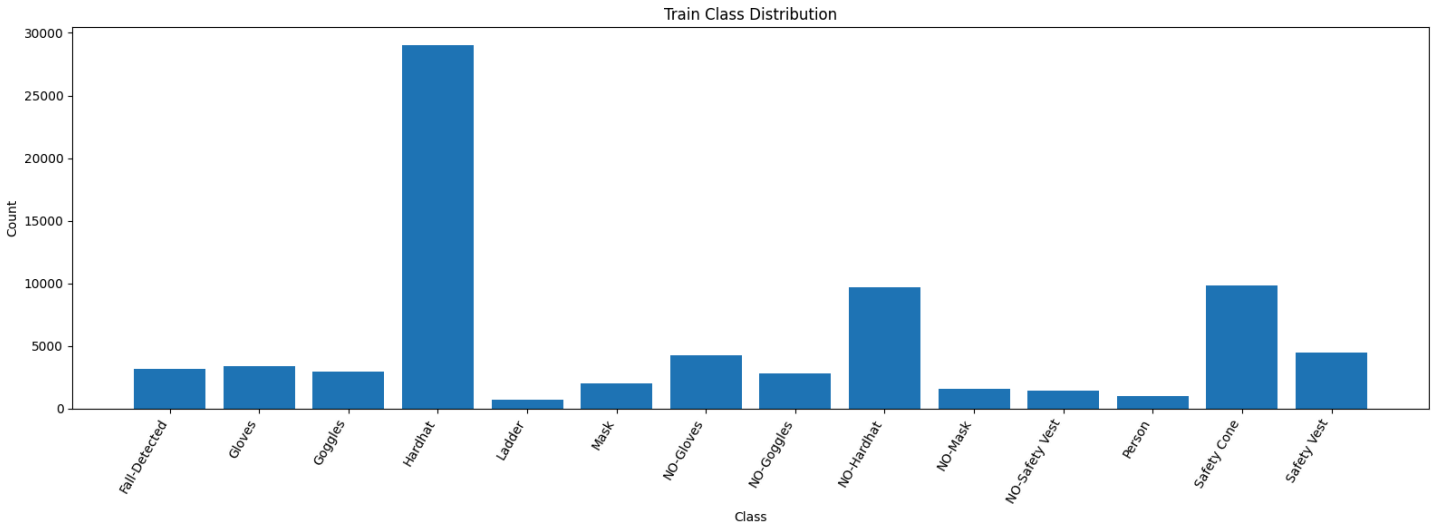

Image4:Training data class distribution

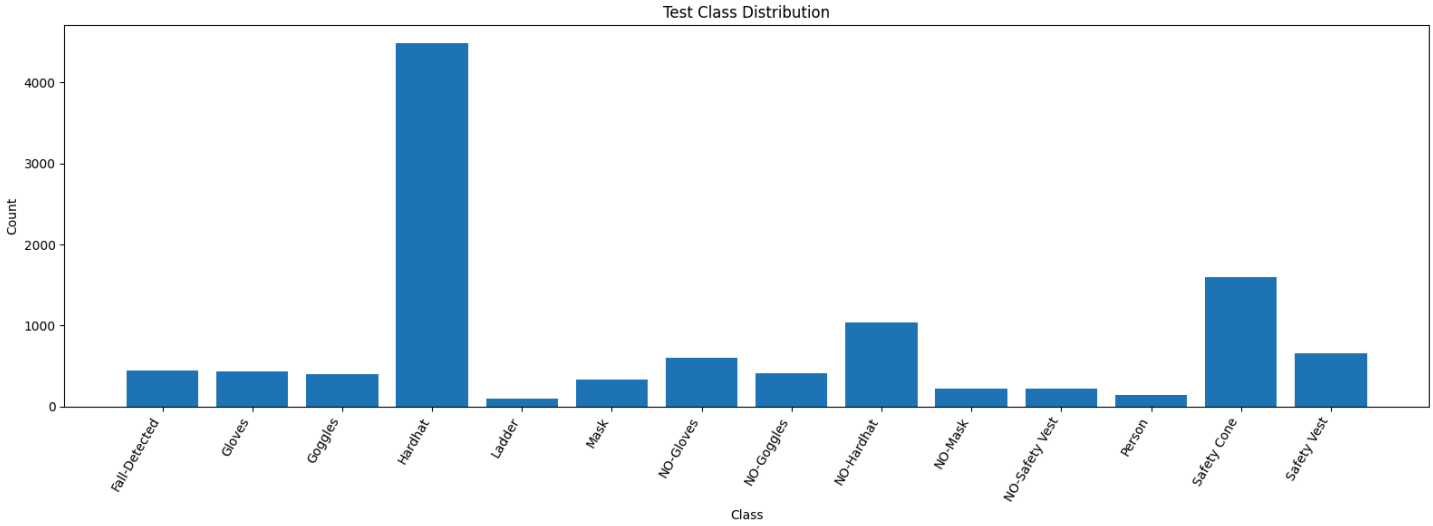

Image5:Testing data class distribution

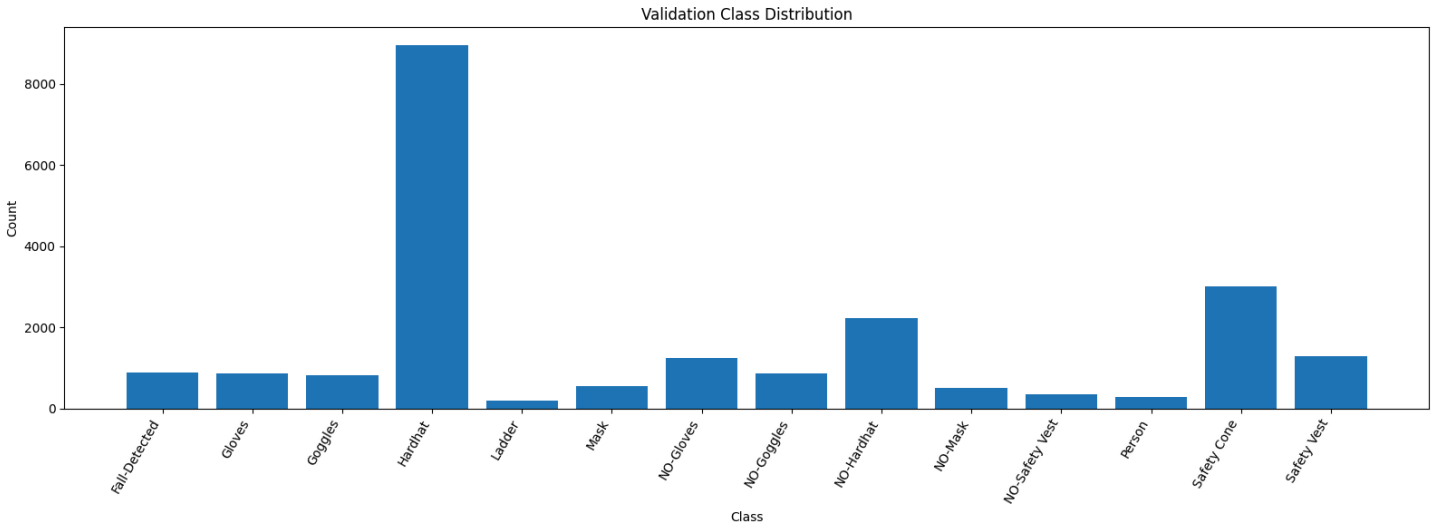

Image6:validation data class distribution

Count Images and Labels in Each Split

This cell counts how many image files and label files are present in each dataset split, helping confirm completeness and detect missing files.

```python
from glob import glob
import os

IMAGE_EXTS = ["*.jpg", "*.jpeg", "*.png", "*.bmp", "*.webp", "*.JPG", "*.JPEG", "*.PNG"]

def count_images(folder):
    total = 0
    for ext in IMAGE_EXTS:
        total += len(glob(os.path.join(folder, ext)))
    return total

def count_labels(folder):
    return len(glob(os.path.join(folder, "*.txt")))

print("Train images:", count_images(train_path))
print("Train labels:", count_labels(train_labels_path))
print("Val images  :", count_images(val_path))
print("Val labels  :", count_labels(val_labels_path))
print("Test images :", count_images(test_path))
print("Test labels :", count_labels(test_labels_path))
```

```
Train images: 30765
Train labels: 30765
Val images  : 8814
Val labels  : 8814
Test images : 4423
Test labels : 4423
```

Image7:Main Dataset class distribution

```
===== TRAIN =====
Image files         : 30765
Label files         : 30765
Missing labels      : 0
Missing images      : 0
Sample missing label: []
Sample missing image: []

===== VALID =====
Image files         : 8814
Label files         : 8814
Missing labels      : 0
Missing images      : 0
Sample missing label: []
Sample missing image: []

===== TEST =====
Image files         : 4423
Label files         : 4423
Missing labels      : 0
Missing images      : 0
Sample missing label: []
Sample missing image: []
```

Image8:Main splitt sets data distribution

Standardize Class Name Format

This cell converts the class definitions into a consistent list format, making them easier to use in charts, summaries, and label interpretation.

```python
if isinstance(names, dict):
    class_names = [names[i] for i in sorted(names.keys())]
else:
    class_names = names

print("Ordered class list:")
for i, name in enumerate(class_names):
    print(i, "->", name)
```

```
Ordered class list:
0 -> Fall-Detected
1 -> Gloves
2 -> Goggles
3 -> Hardhat
4 -> Ladder
5 -> Mask
6 -> NO-Gloves
7 -> NO-Goggles
8 -> NO-Hardhat
9 -> NO-Mask
10 -> NO-Safety Vest
11 -> Person
12 -> Safety Cone
13 -> Safety Vest
```

Image9:class distribution of the main dataset